Computer Science & Engineering Department

Arhitectura Sistemelor de Calcul

Curs 12
cs.pub.ro
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University Politehnica of Bucharest

In cursul anterior

- Nevoia de a compara sisteme de calcul
- Cum masuram Performanta?
- Benchmark-uri pentru masini “seriale”
- Benchmark-uri pentru masini “paralele”

Cuprins

- Top 500 List
- Arhitecturi si Aplicatii Relevante
- Trend de dezvoltare

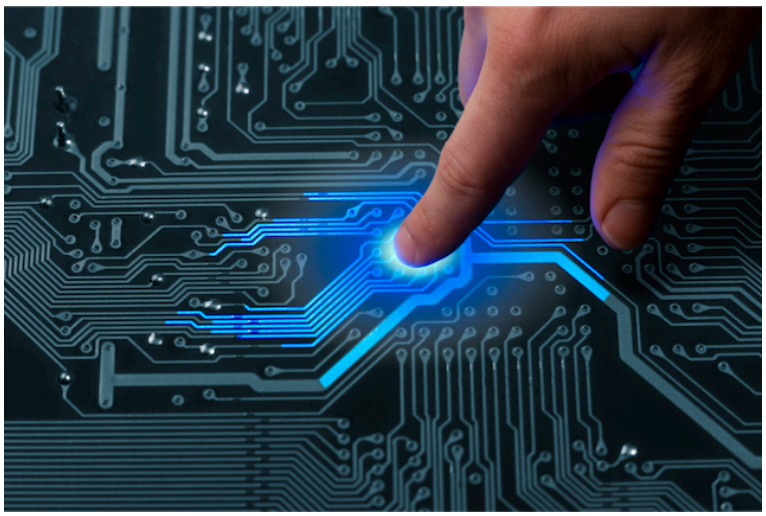
Top 500 List

top500.org

- Proiect demarat in Iunie 1993
- Lista actualizata a celor mai puternice 500 de sisteme de calcul de uz general, disponibile comercial in lume si folosite la aplicatii complexe
- Statisticile referitoare la supercalculatoare sunt interesante pentru
  - Dezvoltatorii de sisteme (producatorii de HW & SW)
  - Utilizatori
  - Potentialii viitori utilizatori
- Se doreste cunoasterea
  - Numarului si locatiei sistemelor
  - Domeniilor de aplicatie ale sistemelor → facilitarea unor potentiale colaborari

top500.org

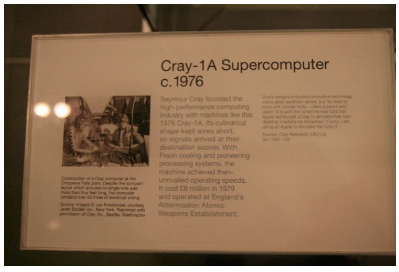
- Lista este publicata de 2 ori pe an
- Benchmark – versiune paralela a Linpack – HPL (Jack Dongarra @ ICL, University of Tennessee)
  - http://www.netlib.org/benchmark/hpl/
  - Rezolvarea unui sistem dens de ecuatii liniare Ax = b
  - Nu reflecta intr-un mod exhaustiv performanta intregului sistem!
  - Dimensiunea problemei poate fi modificata pentru a se potrivi cel mai bine pe arhitectura sistemului masurat
- Listele sunt prezentate la @Supercomputing Conference – US si International Supercomputing Conference – Germania

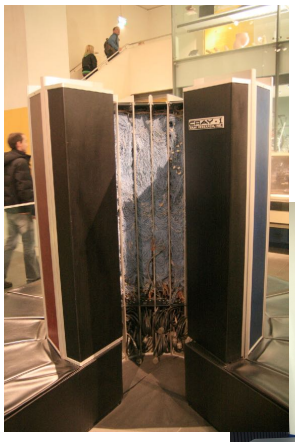


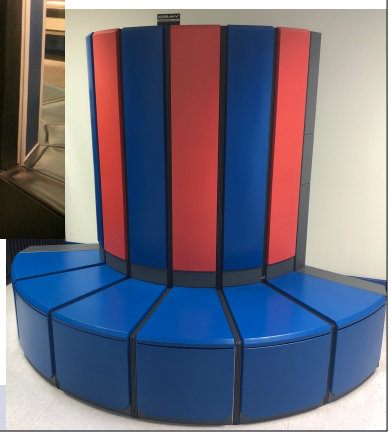
Arhitecturi si Aplicatii Relevante

Motivatie - HW Then…

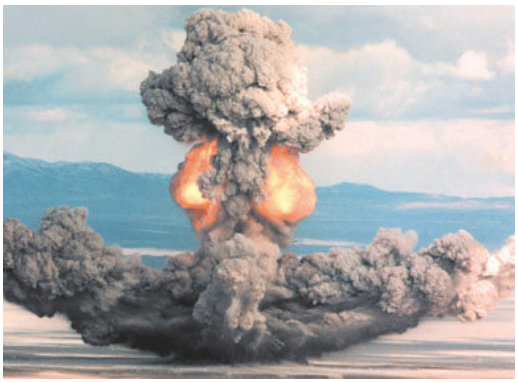
- Cylindrical so wires are short
- Freon Cooling
- 8M Pounds
- Aldermaston Atomic Weapons Establishment







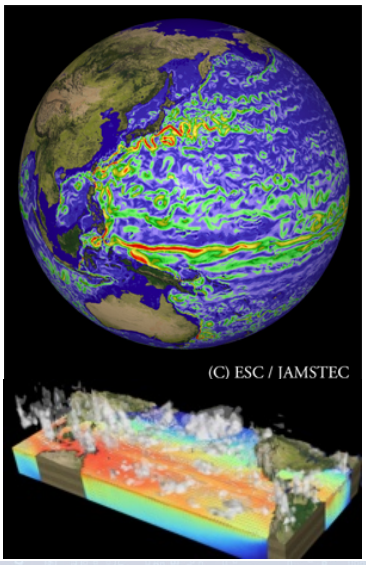
Motivatie - Aplicatii



No (more) Kaboom…

Motivatie - The Earth-Simulator

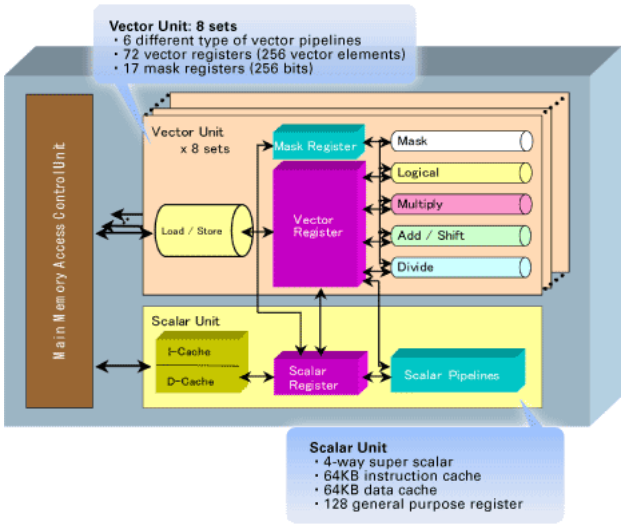
- Simulari atmosferice pentru Agentia Aerospatiala a Japoniei
- Simulari oceanice pentru Centrul Japonez de Stiinte si Tehnologii Marine
- Simulari industriale
- Simulari atomice pentru Insititutul Japonez de **Cercetari Atomice**



Sistemul NEC Earth Simulator

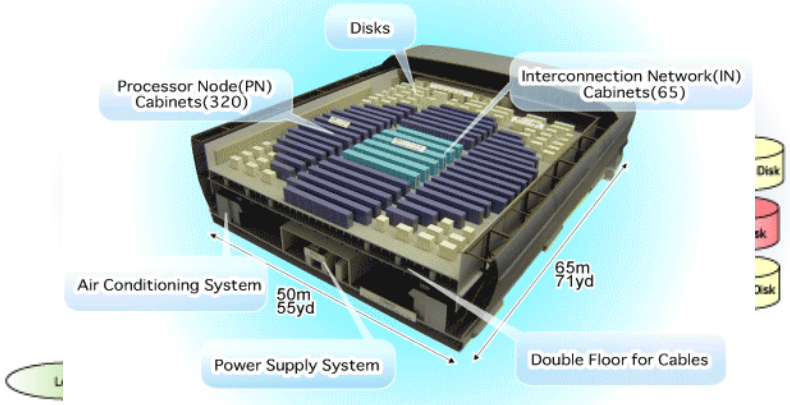
- Site: The Earth Simulator Center
- Familia de sisteme: NEC Vector
- Model: SX6
- Procesor: NEC 1000MHz
- OS: Super-UX
- Arhitectura: MPP
- Aplicatii: Studiul mediului, Cercetari Atomice
- Anul instalarii: 2002
- Numar de procesoare: 5120
- Rmax(TFlops): **35**.86
- Rpeak(TFlops): **40**.96
- Interconectare: Multi-stage crossbar
- No 1 intre 2002-2004

Procesoare Vectoriale SX6



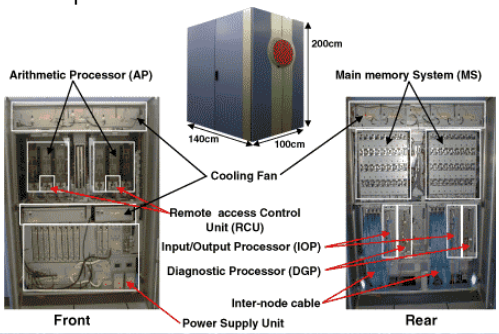
- Unitate Scalara (SU)
- Unitate Vectoriala (VU)
- Frecventa – 500 MHz
- SU – procesor super-scalar: 64Kb instr cache; 64Kb data cache; 128 registrii de uz general
- VU – 72 registrii vectoriali cu 256 elemente & 8 seturi de 6 benzi de asamblare pentru operatii logice si aritmetice
- Tehnologie de fabricaţie: LSI, 0.15µm CMOS, 60 mil. tranzistori
- Consum 140 W

Arhitectura HW Earth-Simulator



Consum & Structura Earth Simulator

- Fiecare 2 noduri sunt instalate într-un container (1x1.4x2m)
- Fiecare container consumă 20 KW → 8MW
- Memoria sistemului este de 10TB
- Spatiu de stocare pe disc 700 TB
- Stocare in masa pe banda 1.6 PB

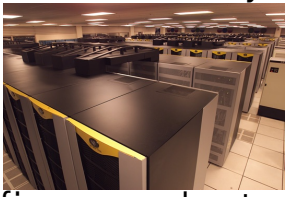


Columbia / SGI Altix @NASA

- Site: NASA/Ames Research Center/NAS
- Familia de sisteme: SGI® Altix™
- Model: SGI® Altix™ 3700, Voltaire Infiniband
- Procesoare: Intel IA-64 Itanium 2 1.5 GHz
- OS: SuSE Linux Enterprise Server 9
- Arhitectura: MPP
- Aplicatii: Cercetari aerospatiale
- Anul instalarii: 2004
- Numar de procesoare: 10240
- Rmax(TFlops): **51**.870
- Rpeak(TFlops): **60**.960
- Interconectare: Numalink/Infiniband

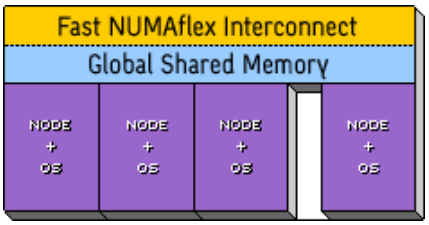
Arhitectura Columbia

- Numele sistemului
  - In memoria echipajului de pe Columbia - 1 februarie 2001
- 20 superclustere Altix™ din seria 3000
  - 8 SGI Altix 3700 si 12 Altix 3700 Bx2
  - 4 din Bx2-uri formeaza un 2048-PE (Processor Shared Memory Environment)
- Procesoare Intel Itanium 2
  - 1.5GHz, 6MB Cache
- Pana la 24 TB de Global Shared Memory pe fiecare cluster

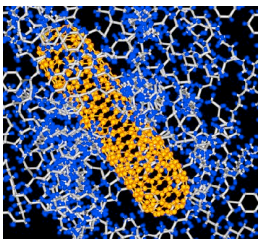


Interconectare Columbia

- Nodurile sunt conectate
  - Voltaire InfiniBand
  - Ethernet de 1 si 10 Gb/s
- Columbia este conectat la un on-line RAID printr-o conexiune Fibre Channel → 440TB
- Conectarea procesoarelor prin SGI®NUMAlink™ → design modular
  - 2048 de procesoara folosesc NumaLink si pentru conectarea dintre noduri
  - Singura tehnologie care ofera global shared memory si intre noduri, nu doar in cadrul unui nod



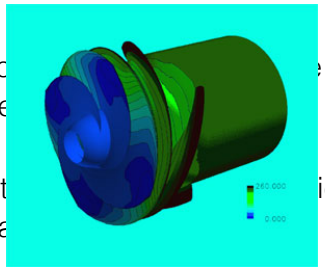
Software Columbia

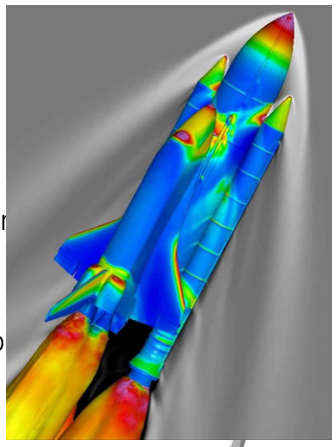


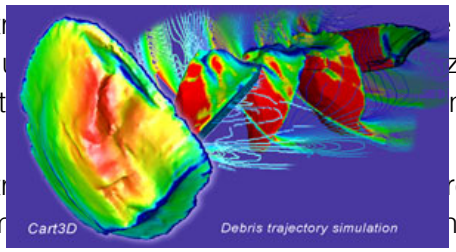
- OS: SGI ProPack 4 – bazat pe SUSE Linux
- Compilatoare Intel® Fortran/C/C++ & Gnu
- Fiecare nod de 512 procesoare ofera
  - Latenţă mică la accesul mem (<1 ms) → reduce overheadul in comunicatie
  - Global shared mem de 1TB → procesele mari rămân rezidente
- Optim pentru aplicatii cu comunicatie masiva intre procesoare
  - Simulari fizice in care domeniul este discretizat → CFD
  - Prognoza meteo & Nanotehnologii
  - N-Body simulations → Astrofizica

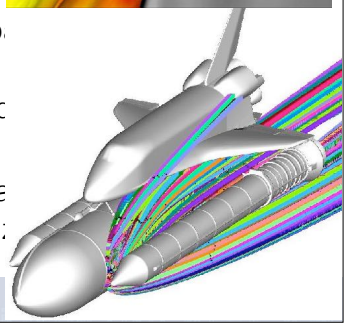
Aplicatii Columbia

- Cart3D
  - Prezice traiectoria unei buc e in timpul ascensiu
  - Culorile reprezinta presiune
- Debris
  - Calculeaza traiectoriile rest icolul orbital in timp
  - Evalueaza periculozitatea a
- Overflow
  - CFD utilizat pentr vehiculelor aerosp
  - Campul de presi za de 2.46Mach, la
  - Culorile reprezint n jurul vehiculului, c
- Phantom
  - CFD utilizat pentr re in motoare cu rea
  - Presiunea in pom navetei spatiale utili







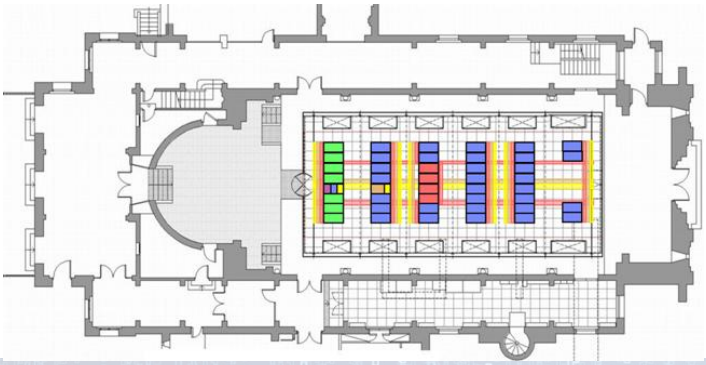


MareNostrum JS20 Cluster

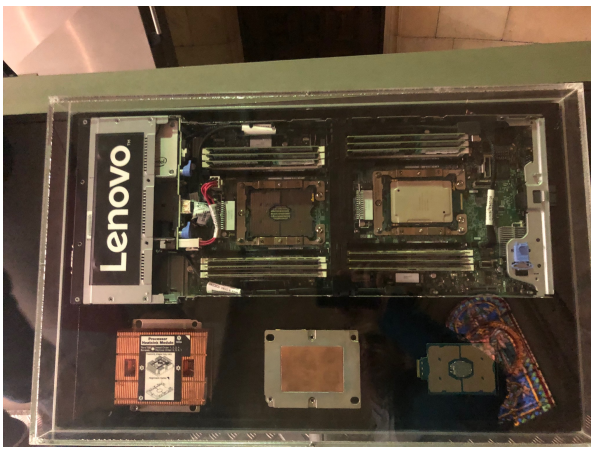
- Site: Barcelona Supercomputing Center
- Familia de sisteme: IBM Cluster
- Model: BladeCenter JS20 Cluster
- Procesoare: PowerPC 970 2.3 GHz
- OS: SuSE Linux Enterprise Server 9
- Arhitectura: Cluster
- Aplicatii: Medicina
- Anul instalarii: 2006
- Numar de procesoare: 10240
- Rmax(TFlops): **62**.63
- Rpeak(TFlops): **94**.21
- Interconectare: Red Myrinet

Arhitectura MareNostrum

- Memorie totala 20TB
- Capacitate de stocare: 370TB
- Este format din 44 de rack-uri ce ocupa 120m^2

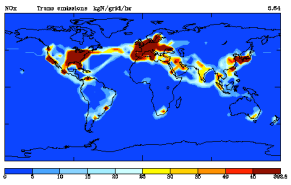


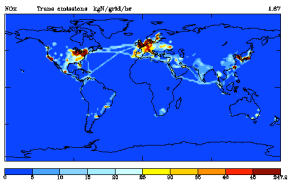
Arhitectura MareNostrum

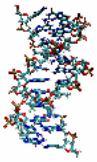


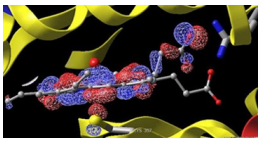
Aplicatii MareNostrum

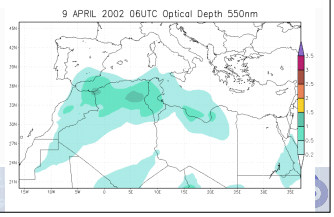
- Computational Sciences
  - Computer Architecture
  - Performance Tools
  - Grid Computing & Clusters
- Earth Sciences
  - Air Quality
  - Meteorological Modelling
  - Mineral Dust
  - Climate Change
- Life Sciences
  - Molecular Modelling & Bioinformatics
  - Computational Genomics
  - Electronic & Atomic Protein Modelling





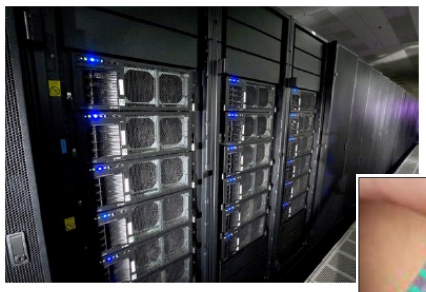


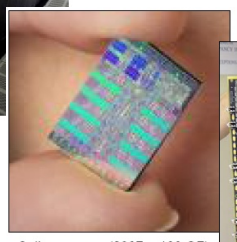




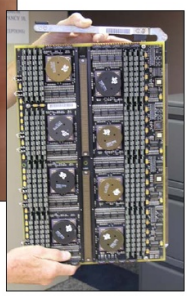
IBM RoadRunner

- Site: DOE/NNSA/Los Alamos NL
- Familia de sisteme:
- Model: BladeCenter QS22 Cluster
- Procesoare: PowerXCell 8i 3.2 GHz
- OS: Linux
- Arhitectura: Cluster
- Memorie: 104TB
- Aplicatii: Cercetare
- Anul instalarii: 2008
- Numar de procesoare: 122.400
- Rmax(PFlops): **1**,042 (NMax 2,25M)
- Rpeak(PFlops): **1**,376
- Consum: 2345 KW
- Interconectarea: Voltaire Infiniband



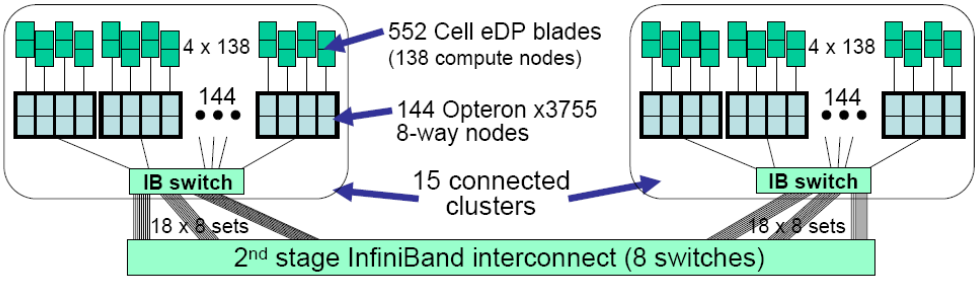


Cell processor (2007, ~100 GF)



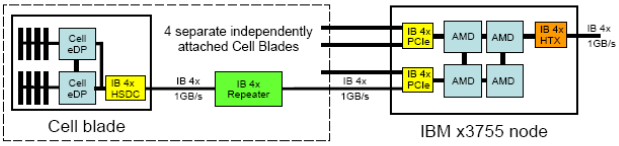
CM-5 board with vector units (1994, 1 GF)

Arhitectura RoadRunner



8,640 dual-core Opterons
⇨ **76 TeraFlops**
16,560 Cell eDP chips
⇨ **1.7 PetaFlops**

One Accelerated Node

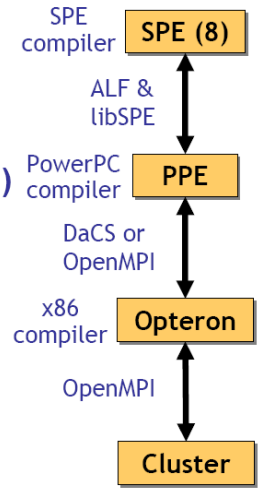


1 Cell processor per Opteron core

Los Alamos NATIONAL LABORATORY

Programare RoadRunner

- **Computational Library (ALF w/ IBM)**
  - Task & work block queuing & management
  - Streaming & user-defined data partitioning
  - Process management
  - Error handling
- **Communication Library (DaCS w/ IBM)**
  - Data movement & synchronization
  - Process management & synchronization
  - Topology description
  - Error handling
  - First implementation may leverage OpenMPI
- **Longer term**
  - ALF & DaCS support in tools
  - ALF from Opteron ⇒ Cell directly
  - Compilers supporting some of this



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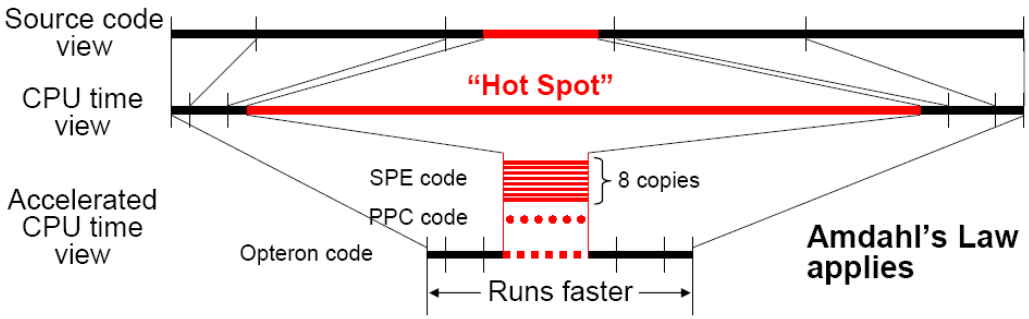
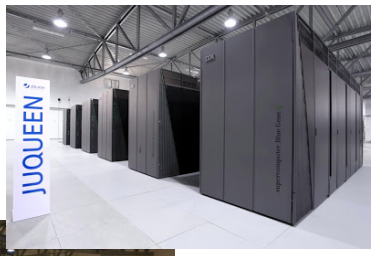
Programare Hibrida RoadRunner



- Hybrid programming will be a challenge!
  - No compiler switches to "just use the Cells"
  - Not even a single compiler – 3 of them
  - Code developer/architect must decompose application and create cooperative program pieces

JuQueen BlueGene/Q

- Site: Forschungszentrum Juelich (FZJ)
- Model: IBM-BG/Q – 28 Racks
- Procesor: IBM PowerA2, 1.6 GHz
- OS: CNK/Linux
- Arhitectura: MPP
- Aplicatii: Cercetare
- Anul instalarii: 2012
- Numar de core-uri: 458.752
- Rmax(PFlops): **5**,009
- Rpeak(PFlops): **5**,872
- Memorie: 448 TB
- Interconectare: Custom 5D Torus



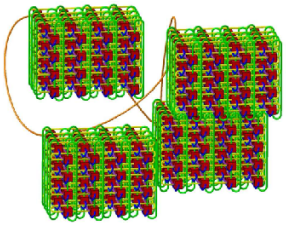
ASC@UPB: Monica Bugeanu & Alex Calotoiu

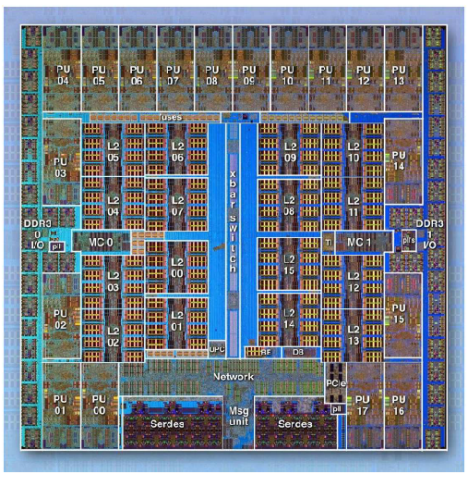
Arhitectura IBM BlueGene

- Program initiat de IBM in 1999 pentru a construi “a petaflop scale machine”
- BlueGene/L – primul pas, bazat pe procesoare PowerPC
  - Spatiu de adresare mare
  - Compilatoare standard
  - Bazat pe middleware de “message passing” deja existent
  - A necesitat adaugiri semnificative fata de sistemul PowerPC standard
- Un nod computational = computer-on-a-chip – ASIC
  - Procesoare IBM PowerPC
  - Memorie DRAM embedded
  - L1-3 cache embedded
  - Multiple module de interconectare folosind retele de comutare de mare viteza

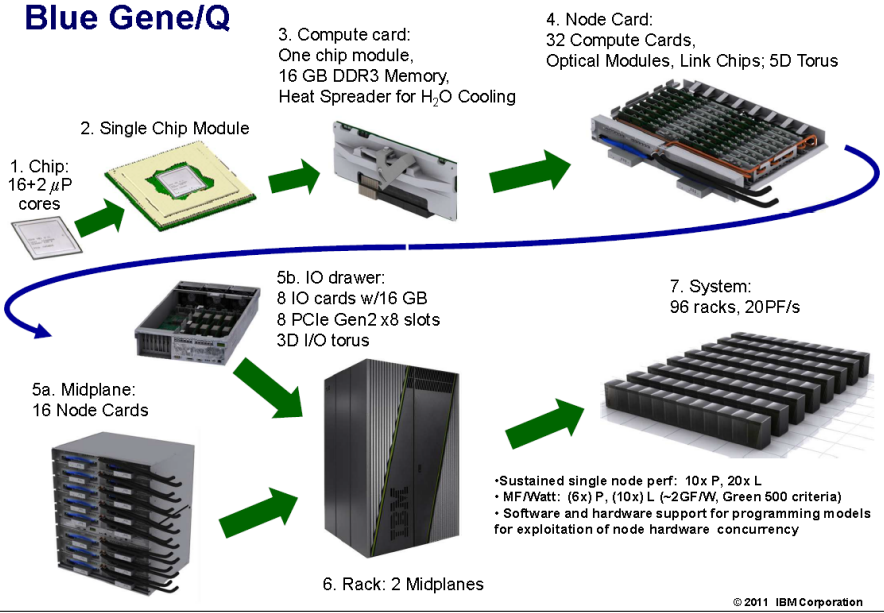
Arhitectura IBM BlueGene

- Memorie
  - 1GB DDR RAM / core – 16GB pe nod
  - Doua controlere de memorie externa de tip DDR integrat on-chip la 1.33GHz
- Procesor
  - 18C @ 1.6GHz: 16 computationale, unul pentru OS, unul de backup
  - 32MB DRAM L2 Cache: transactional memory & executie speculativa
  - Quad SIMD floating point unit
  - 45nm cu performanta maxima de 204.8 GFLOPS
  - Consuma 55W la 19.19mm si 1.47 B tranzistoare
- Interconectare
  - Tor 5D cu un router pe nod cu 11 link-uri
  - Procesor I/O extern dedicat – 2GB/s
  - Core-urile legate printr-un crossbar
- I/O Extern
  - Noduri dedicate pentru I/O extern
  - Reteaua este de tip arborescent
- OS - CNK/Linux
- Software: MPI, OpenMP, Charm++, GlobalArrays, Posix IO





Arhitectura IBM BlueGene



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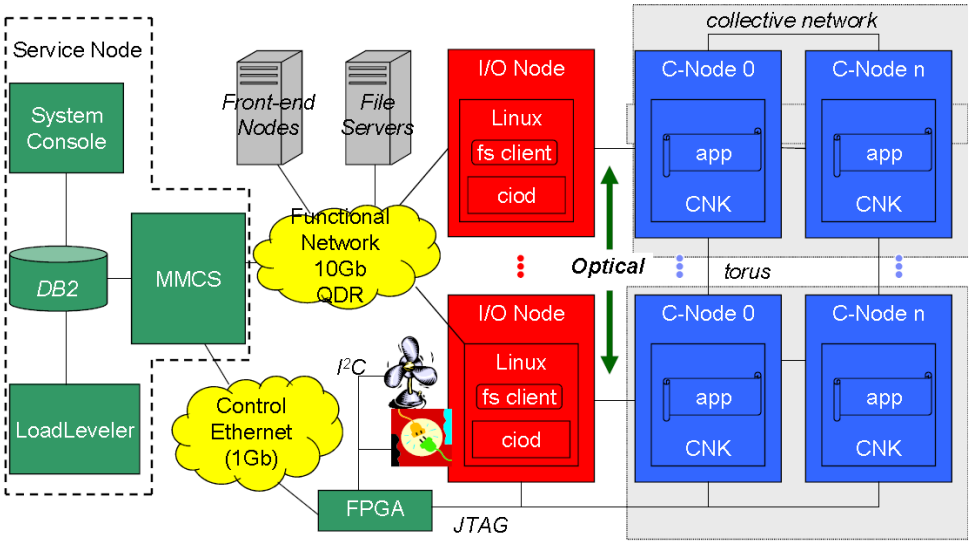
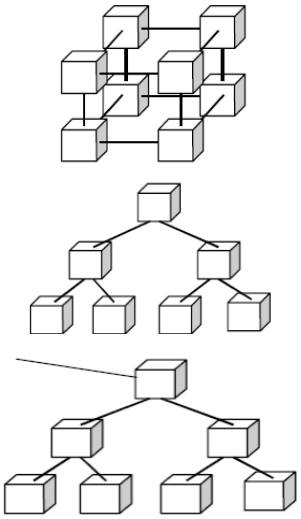
Arhitectura IBM BlueGene

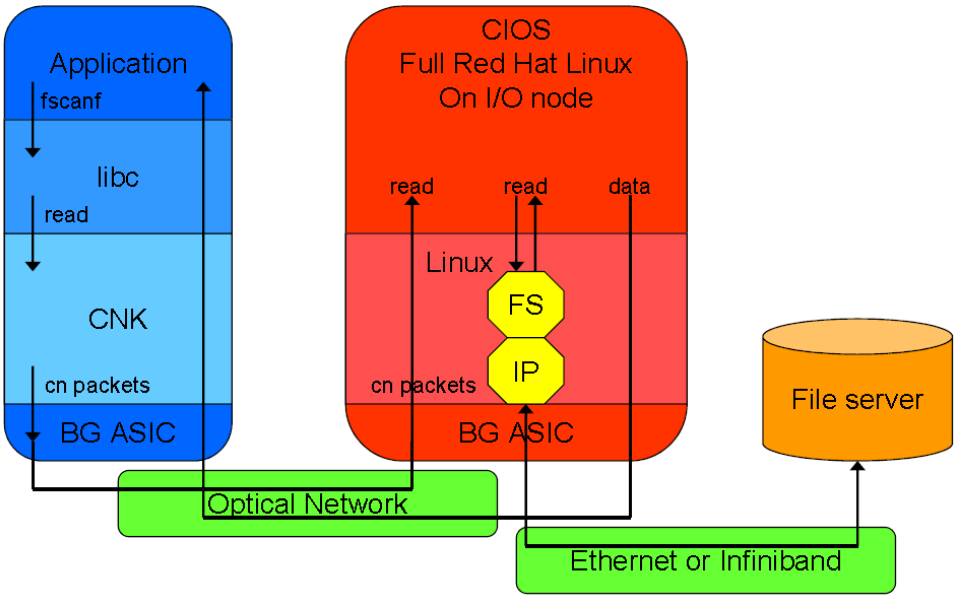


Retele de Comunicatie IBM BlueGene

- Nodurile sunt atasate la 5 retele de comunicatie
  - Retea toroidala 5D pentru comunicatii intre noduri
  - Retea colectiva de comunicatii
  - Retea globala de intreruperi si bariere
  - Pentru I/O – Gigabit Ethernet
  - Joint Test Access GroupGigabit Ethernet – pentru control si monitorizare
- Favorizeaza aplicatii care comunica cu vecinii imediati
- BG e dezvoltat pentru aplicatii cu volum mare de date



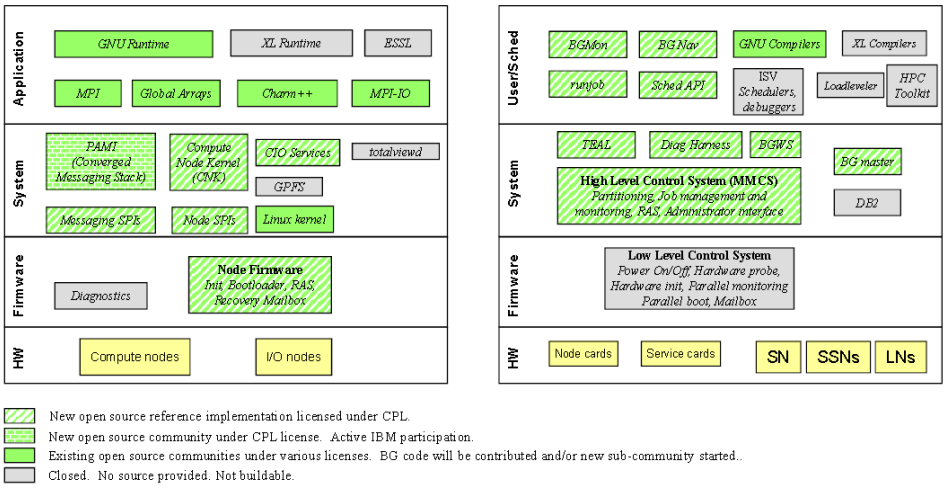
I/O IBM BlueGene



IBM BlueGene Software Stack

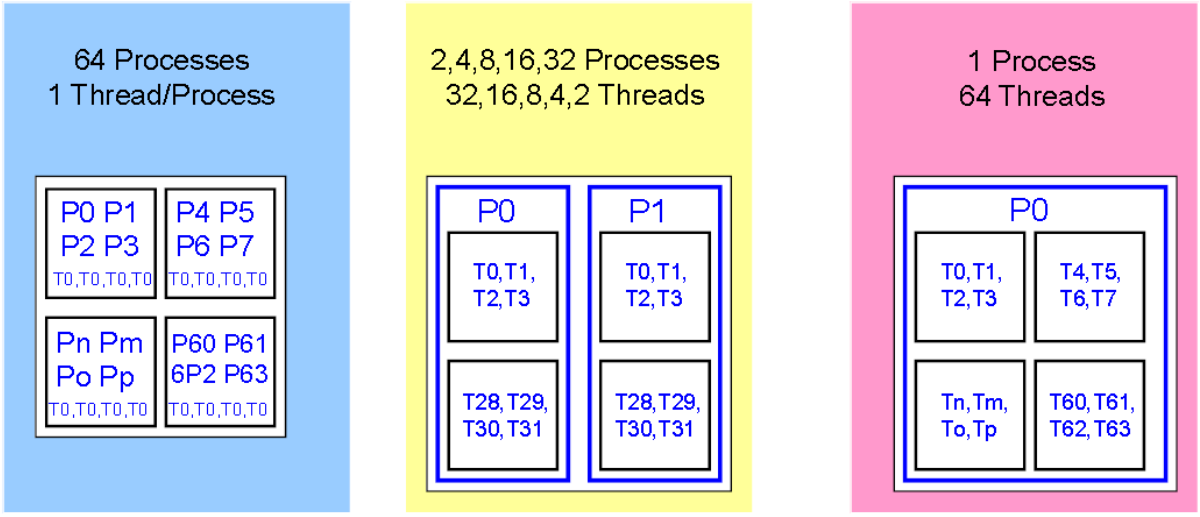
I/O and Compute Nodes

Service Nodes/Login Nodes



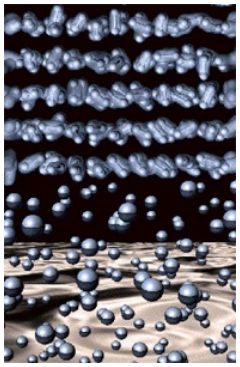
New open source reference implementation licensed under CPL.
New open source community under CPL license. Active IBM participation.
Existing open source communities under various licenses. BG code will be contributed and/or new sub-community started..
Closed. No source provided. Not buildable.

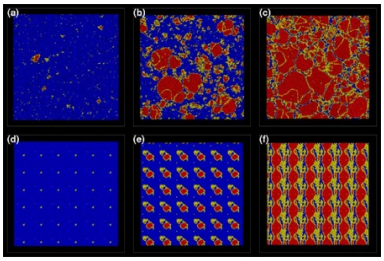
IBM BlueGene – Mod de executie



Aplicatii IBM BlueGene

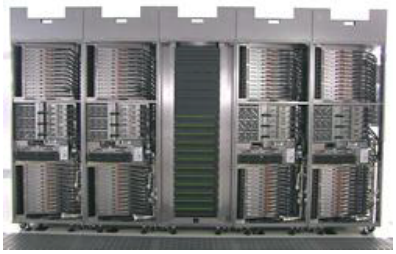
- Analiza proteinelor
- Interactiunea intre medicamente si proteine
- Data Mining
- Fizica atomica: similar cu ASC Purple
- Dinamica moleculara (inclusiv ab-initio)
- Hidrodinamica steady state si turbulenta
- Astrofizica

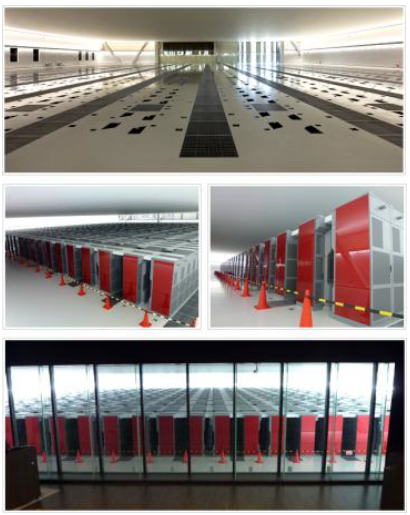




K Computer SPARC64

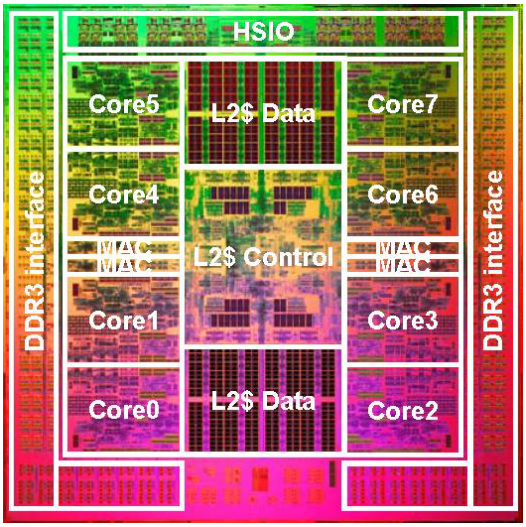
- Site: RIKEN Advanced Institute for Computational Science
- Familia de sisteme: Fujistu – SPARC64 VIIIfx 2.0GHz
- Procesor: SPARC64 VIIIfx 2.0GHz (8-core)
- OS: Linux
- Arhitectura: Cluster
- Aplicatii: Cercetare
- Anul instalarii: 2011
- Numar de core-uri: 705.024
- Rmax(PFlops): **10**,510
- Rpeak(PFlops): **11**,280
- HPCG(TFlops): **602**,5
- Memorie: 1,410 TB
- Consum: 12.660 kW
- Interconectare: Custom – Tofu



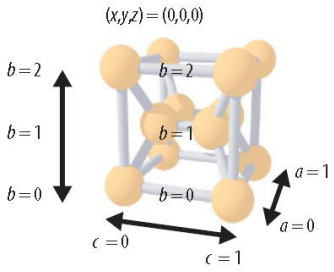


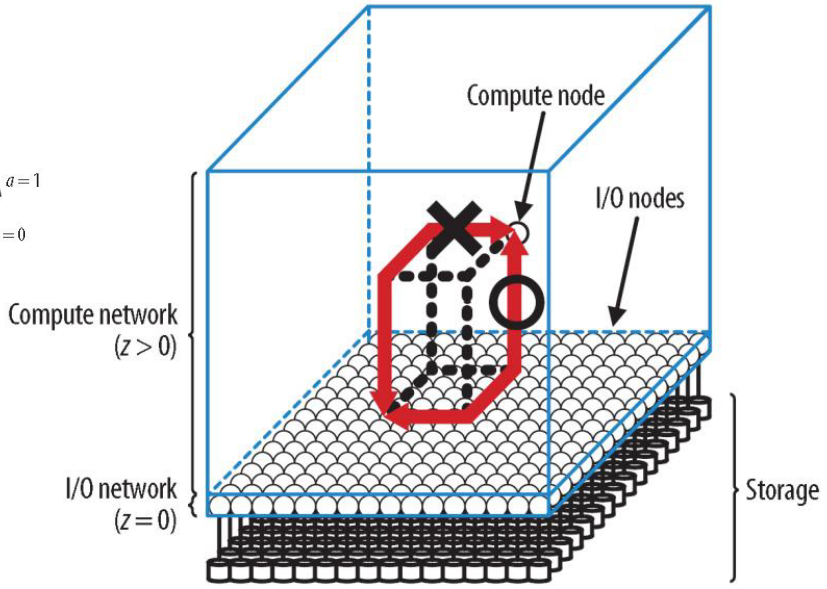
K Computer Processor

- 8C @ 2GHz – SoC design
- Embedded Memory Controller
- Shared 5M L2 Cache
- 45nm – 760M tranzistori
- 128GFlops (peak)
- 64GB/s transfer la memorie
- 58W si racire cu apa
- Set de registri mare 192 int / 256 fp
- Unitati SIMD
- Software Managed Cache



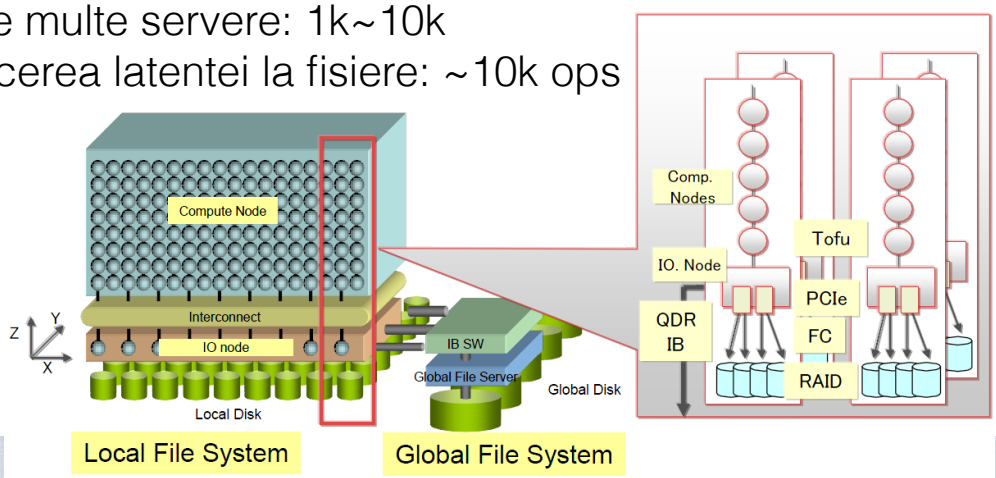
K Computer 6D Mesh/Torus Tofu Interconnect



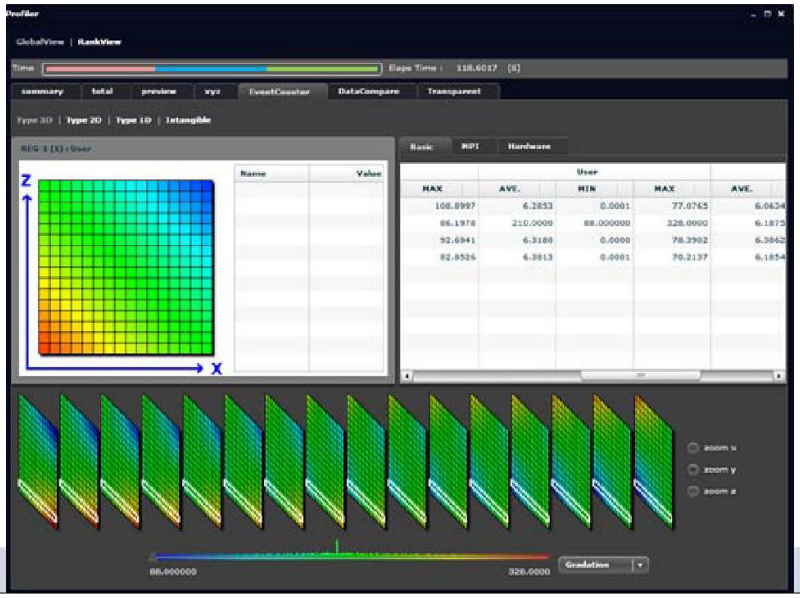


K Computer Fujitsu Exabyte File System

- Date foarte multe: 100PB~1EB
- Foarte multi clienti: 100k~1M
- Foarte multe servere: 1k~10k
- Reducerea latentei la fisiere: ~10k ops



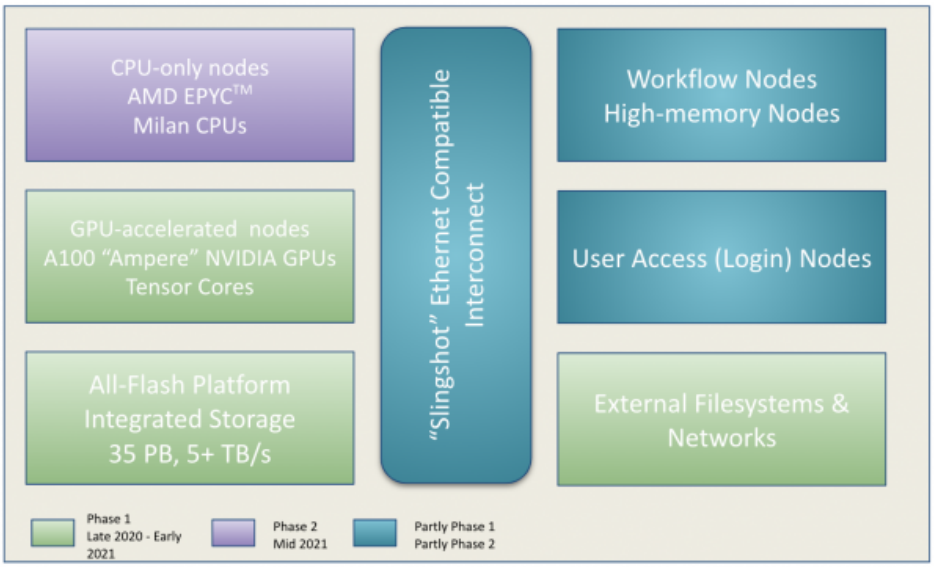
K Computer Performance Tuning



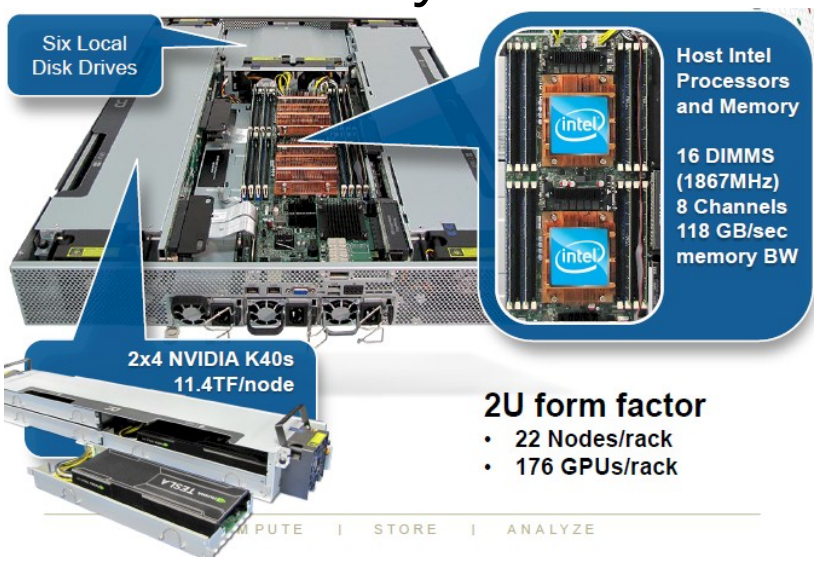
14/2024 - Perlmutter

- Site: LBNL
- Familia de sisteme: HPE Cray EX235n
- Procesor: AMD EPYC 7763 64C 2.45GHz, NVIDIA A100 SXM4 40 GB
- OS: HPE Cray OS
- Arhitectura: MPP
- Aplicatii: Research
- Anul instalarii: 2021
- Numar de core-uri: 761,865
- Rmax(PFlops): **70**,90
- Rpeak(PFlops): **93**,75
- HPCG(TFlops): **1,905**
- Memorie: **421** TB
- Consum: **2.58** kW / **27.37** GFlop/W
- Interconectare: GBEth / Slingshot-10 (274 GB)

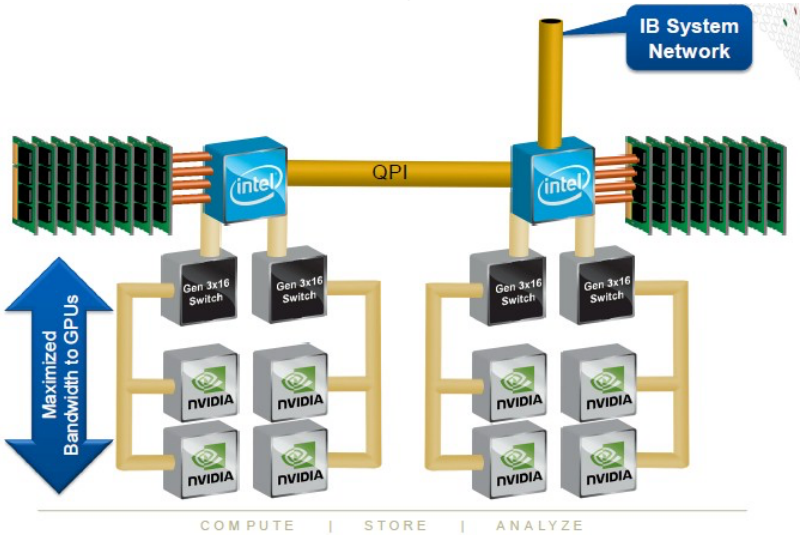
Perlmutter Architecture



Arhitectura Cray XC Series – 1



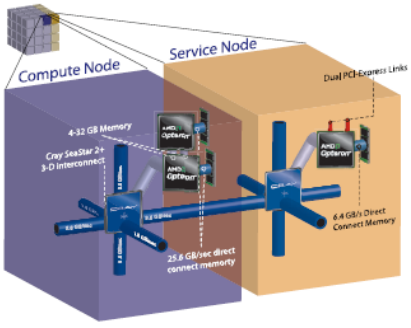
Arhitectura Cray XC Series – 2

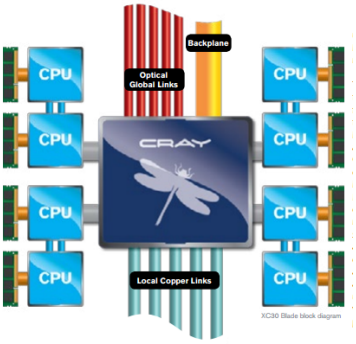


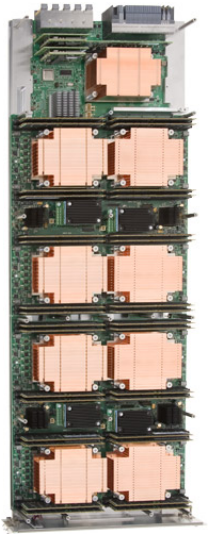
Cray XC – Software Stack

Cray HPC Cluster Software Stack						
HPC Programming Tools	Development & Performance Tools	Cray PE on CS	Intel® Cluster Studio	PGI Cluster Development Kit®	GNU Toolchain	NVIDIA® CUDA®
	Application Libraries	Cray® LibSci™, LibSci_ACC	Intel® MPI	IBM Platform MPI	MVAPICH2	OpenMPI
	Debuggers	Rogue Wave TotalView®	Allinea DDT, MAP	Intel® IDB	PGI PGDBG®	GNU GDB
Schedulers File Systems and Management	Resource Management / Job Scheduling	SLURM	Adaptive Computing Moab®, Maui, Torque	Altair PBS Professional	IBM Platform™ LSF®	Grid Engine
	File Systems	Lustre®	NFS	GPFS	Panasas PanFS®	Local (ext3, ext4, XFS)
	Cluster Management	Cray® Advanced Cluster Engine (ACE™) Management Software				
Operating Systems and Drivers	Drivers & Network Mgmt.	Accelerator Software Stack & Drivers		OFED		
	Operating Systems	Linux® (Red Hat, CentOS)				

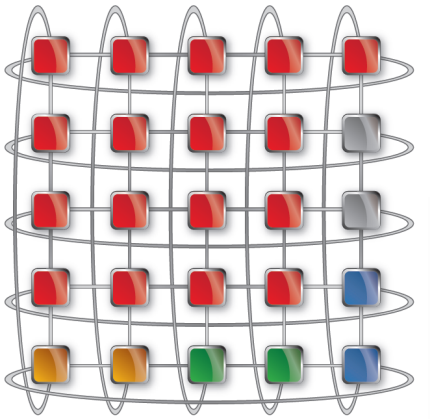
Arhitectura Sisteme Cray

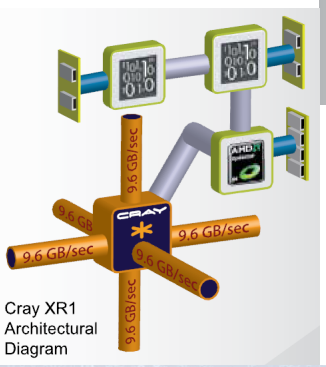




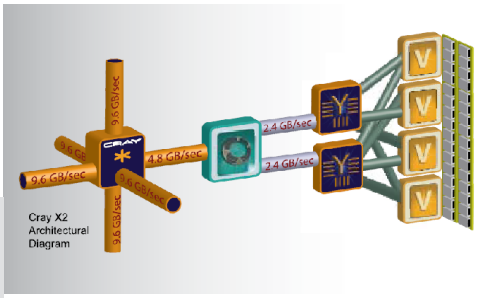


Arhitectura Sisteme Cray



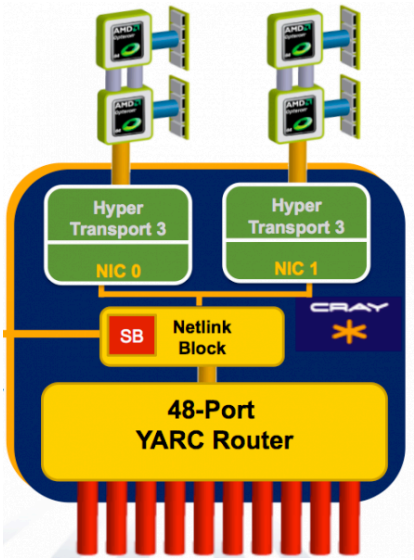


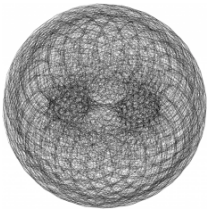
Cray XR1
Architectural
Diagram



Cray X2
Architectural
Diagram

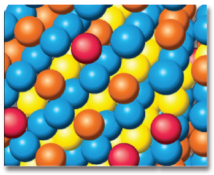
Retele de conectare Cray

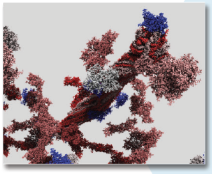




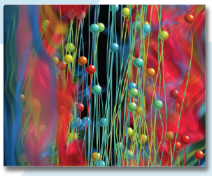
Cray - Aplicatii

WL-LSMS
Role of material disorder, statistics, and fluctuations in nanoscale materials and systems.



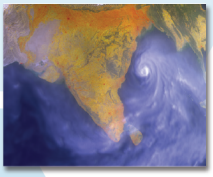


LAMMPS
A multiple capability molecular dynamics code.

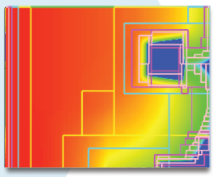


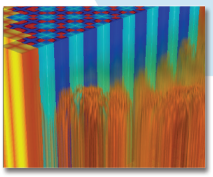
S3D
Combustion simulations to enable the next generation of diesel/bio fuels to burn more efficiently.

CAM-SE
Answers questions about specific climate change adaptation and mitigation scenarios.



NRDF
Radiation transport – important in astrophysics, laser fusion, combustion, atmospheric dynamics, and medical imaging – computed on AMR grids.



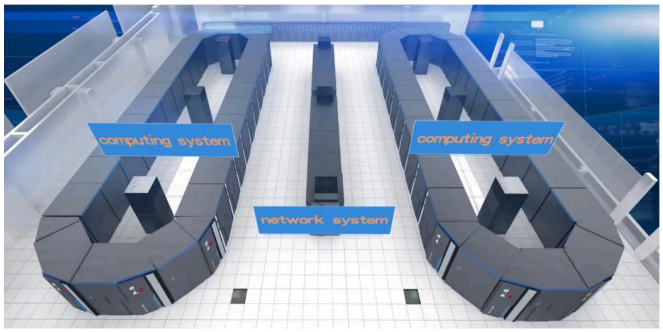


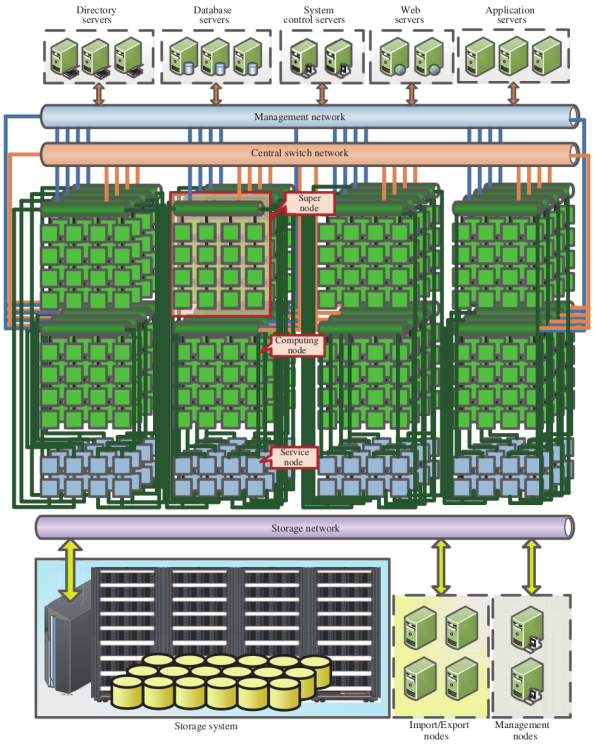
Denovo
High-fidelity radiation transport calculations that can be used in a variety of nuclear energy and technology applications.

13/2024 - Sunway TaihuLight

- Site: National Super Computer Center in Wuxi
- Model: NRCPC MPP
- Procesor: Sunway SW26010 260C 1.45GHz
- OS: Sunway RaiseOS 2.0.5
- Arhitectura: MPP
- Aplicatii: Cercetare
- Anul instalarii: 2016-2018
- Numar de core-uri: 10.649.600
- Rmax(PFlops): **93**,02
- Rpeak(PFlops): **125**,44
- HPCG(TFlops): **481**
- Memorie: **1,311** TB
- Consum: **15,371** kW
- Interconectare: Custom Sunway

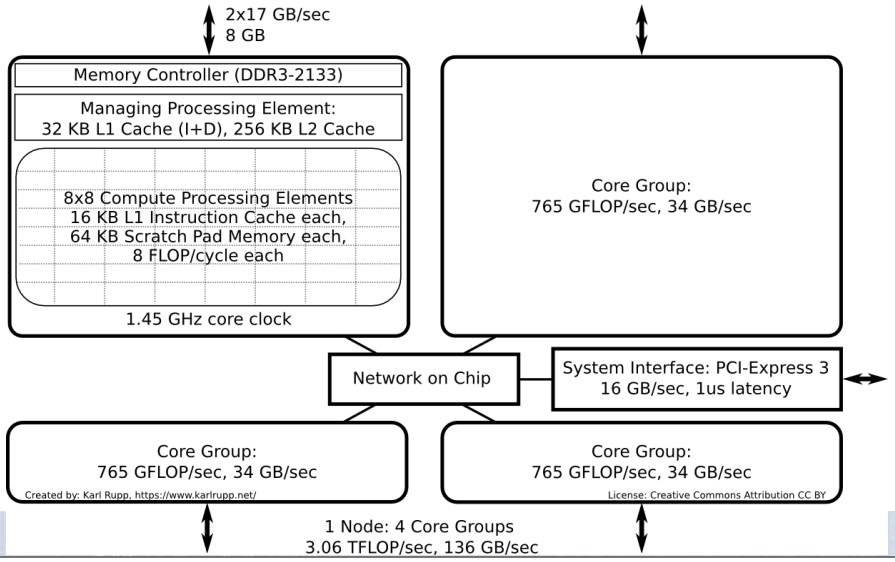
Arhitectura Sunway TaihuLight





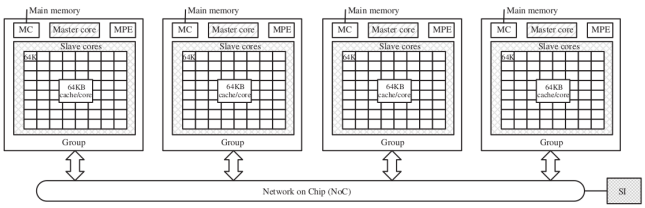
Node Architecture Sunway TaihuLight

1 Node (of 40 960) of Sunway TaihuLight

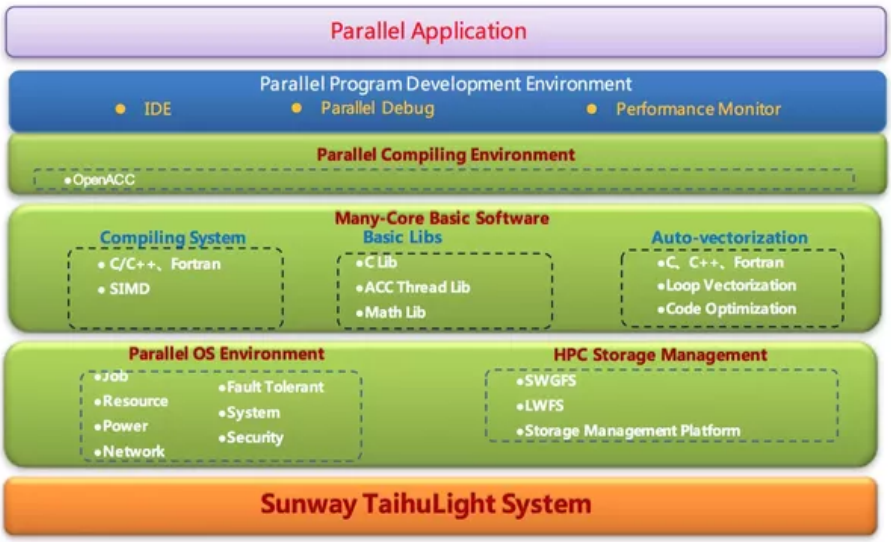


Arhitectura Sunway TaihuLight

- Performance / Watt: 6.05 GFLOPS/W - 4th Top500 Green
- 40960 Processors, each with 260 Cores having:
  - Scratchpad Memory: 64kB Data / 16kB Instruction
- Memory Bandwith: 5591 TB/s
- Network link Bandwith: 14GB/s
- System Storage: 20PB
- I/O Bandwidth: 288GB/s
- NoC Interconnect



Software Stack Sunway TaihuLight



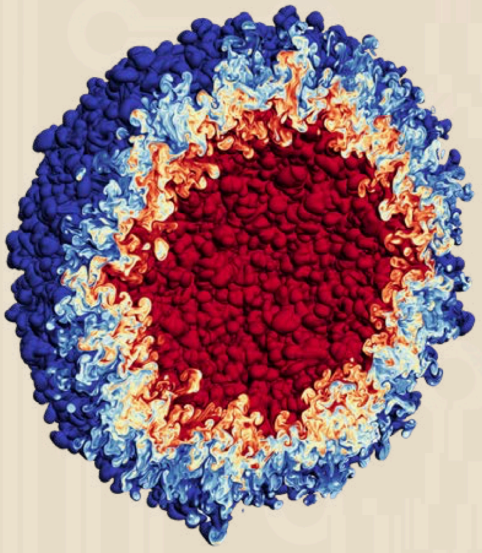
Software Sunway TaihuLight

- Simulation/Modeling Packages:
  - ANSYS: Fluent, CFX, Multiphysics
  - Comet
  - Star CD / CCM+
  - MSC Nastran
  - LS-Dyna
- Pre/post processing utilities:
  - HyperMesh
  - Tecplot
  - Gambit
  - MSC Patran

12/2024 - Sierra Power System S922

- Site: DOE/NNSA/LLNL
- Familia de sisteme: IBM Power Systems
- Procesor: IBM POWER9 22C 3.1GHz, NVIDIA Volta V100
- OS: Red Hat Enterprise Linux
- Arhitectura: Cluster
- Aplicatii: Cercetare
- Anul instalarii: 2018
- Numar de core-uri: 1,572,480
- Rmax(PFlops): **94**,64
- Rpeak(PFlops): **125**,71
- HPCG(PFlops): **1,796**
- Memorie: **1.382** TB
- Consum: **7.438** kW
- Interconectare: Dual-rail Mellanox EDR Infiniband

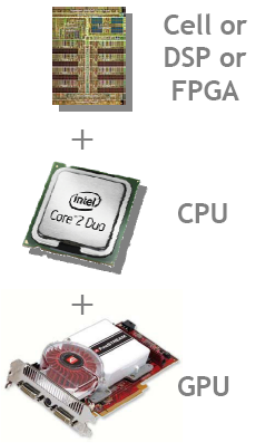
Application Highlight



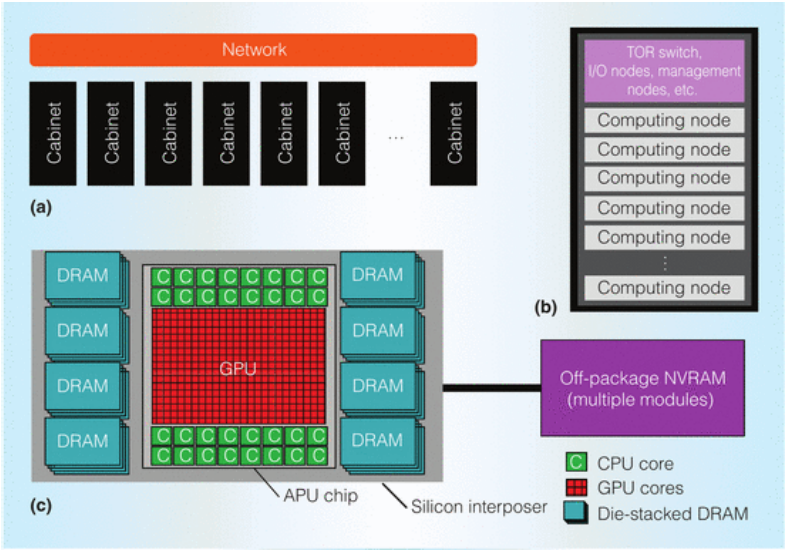
- Nuclear Security Administration's stockpile stewardship mission
- 3D Simulation instead of underground testing
- Performance of nuclear weapon systems
- Understand key issues of nuclear physics

Future Trends Processors

- New processors & accelerators:
  - Multi-core to many-core: 8, 16, 32, ... 80, 128, 1000?
  - IBM Cell , AMD Fusion, Intel Polaris, NVidia G8800
  - Distributed memory & caches at core level
  - FPGAs, GPGPU, Clearspeed CSX600, IBM Cell, XtremeData XD1000, Nvidia G80, AMD Stream Processor
- Connection standards
  - AMD Torrenza, Intel/IBM Geneseo, AMD HyperTransport Initiative
- Programming
  - IBM Roadrunner Cell libraries, RapidMind, Peakstream, Impulse C, Standford's Sequoia, NVidia CUDA, Clearspeed C, Mercury MFC, stream programming
- Heterogeneous architectures
  - Clusters of mixed node
  - Hybrid accelerated node (e.g. Roadrunner, Clearspeed, FPGA)
  - Hybrid on the same bus (e.g. CPU+GPUs, Intel's Geneseo, AMD's Torrenza)
  - Within processor itself (e.g. Cell, AMD Fusion)



Future Trends System Architecture



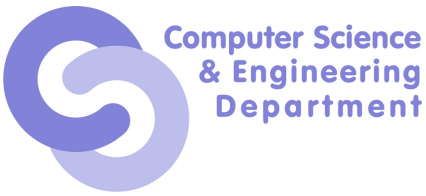
Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,206.00	1,714.81	22,786
2	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
3	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
4	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
5	**LUMI** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107

The Current Top 5 @Top 500

Ce am invatat in aces curs?

- Top 500 List
- Arhitecturi si Aplicatii Relevante
- Trend de dezvoltare

Thank you for your attention.



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